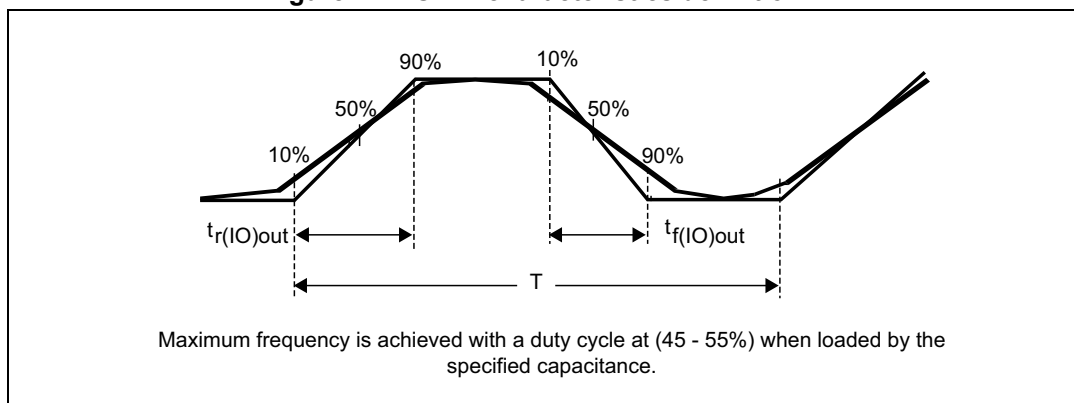


Figure 21. I/O AC characteristics definition⁽¹⁾

1. Refer to [Table 52: I/O AC characteristics](#).

5.3.14 NRST input characteristics

The NRST input driver uses CMOS technology. It is connected to a permanent pull-up resistor, R_{PU} .

Unless otherwise specified, the parameters given in the following table are derived from tests performed under the ambient temperature and supply voltage conditions summarized in [Table 24: General operating conditions](#).

Table 53. NRST pin characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{IL(NRST)}$	NRST input low level voltage	-	-	-	$0.3 \times V_{DD}$	V
$V_{IH(NRST)}$	NRST input high level voltage	-	$0.7 \times V_{DD}$	-	-	V
$V_{hys(NRST)}$	NRST Schmitt trigger voltage hysteresis	-	-	200	-	mV
$R_{PU}^{(1)}$	Weak pull-up equivalent resistor ⁽²⁾	$V_{IN} = V_{SS}$	25	40	55	k Ω
$V_{F(NRST)}^{(1)}$	NRST input filtered pulse	$2.0\text{ V} < V_{DD} < 3.6\text{ V}$	-	-	70	ns
$V_{NF(NRST)}^{(1)}$	NRST input not filtered pulse	$2.0\text{ V} < V_{DD} < 3.6\text{ V}$	350	-	-	ns

1. Specified by design. Not tested in production..

2. The pull-up is designed with a true resistance in series with a switchable PMOS. This PMOS contribution to the series resistance is minimal (~10% order).